

Pricing: a Monte Carlo search for the fair yield

The fair yield is the annual coupon that makes the note worth exactly USD 100 today, using market volatilities and correlations

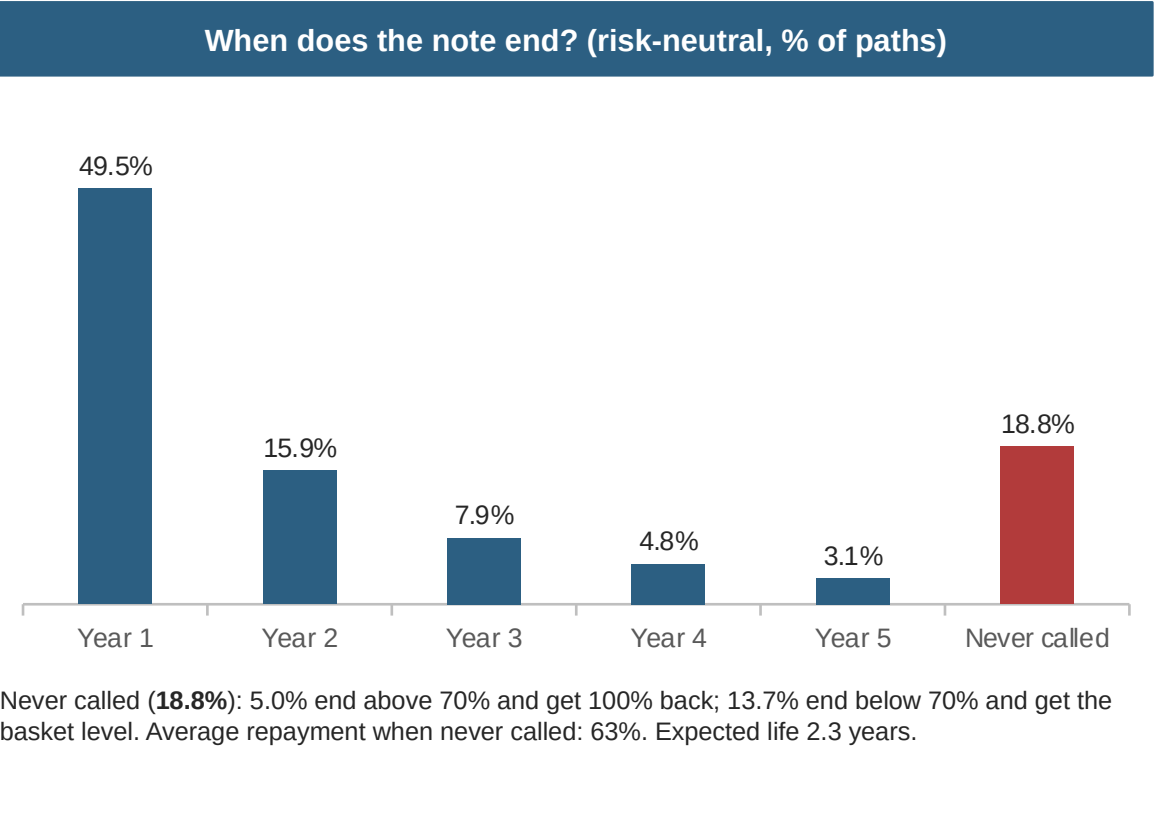
1 The note we price		2 Market inputs						3 Method	
Observation	Annual, Years 1–5	Implied vol	1Y	2Y	3Y	4Y	5Y	1	Simulate: 1,000,000 correlated lognormal basket paths (seed 12345), risk-neutral drift = 5.69%
Autocall trigger	102.5% → 100% → 97.5% → 95% → 92.5% (steps down 2.5% a year)	3119 HK	30%	32%	34%	37%	38%		
		EWY	42%	43%	45%	45%	45%		
		EWT	32%	36%	36%	36%	36%		
If called in year k	USD 100 × (1 + k × yield): the coupon accrues, paid at the call	2644 JP	23%	25%	26%	26%	26%	2	Record: the first year the basket is at or above that year's trigger, or 'never called'
Never called	Year 5: 100% back if basket ≥ 70%; below 70%, the basket level	00878 TT	7%	9%	12%	12%	12%		
Basket	3119 HK 35%, EWY 20%, EWT 15%, 2644 JP 15%, 00878 TT 15% (USD)	Term structure used properly: each year's variance is the increase in total variance ($t \times \sigma^2$), not the quoted vol itself. Correlations: supplied 5×5 matrix (0.44–0.86), checked positive definite (smallest eigenvalue 0.11). Rate: 5.69% case rate, annually compounded; no dividends (USD performance assumed).						3	Value: every path's cash flows, discounted: $PV = E[a] + \text{yield} \times E[b]$
Discounting	5.69% a year (case USD rate), flat							4	Solve: $\text{yield} = (1 - E[a]) / E[b]$, so the note is worth exactly 100%

a = discounted principal repaid, b = discounted years of coupon earned: the fair yield balances the two at USD 100

Source: team pricing brief and stochastic_cal_for_natixis.py. Excludes issuer credit, fees and hedging costs. Vol inputs: USD vs local-currency basis to confirm; 2644 JP row to confirm (originally labelled 2664).

Result: the fair yield is 13.81% a year

1,000,000 simulated paths; we take the bottom of the 95% confidence interval so the yield is never overstated



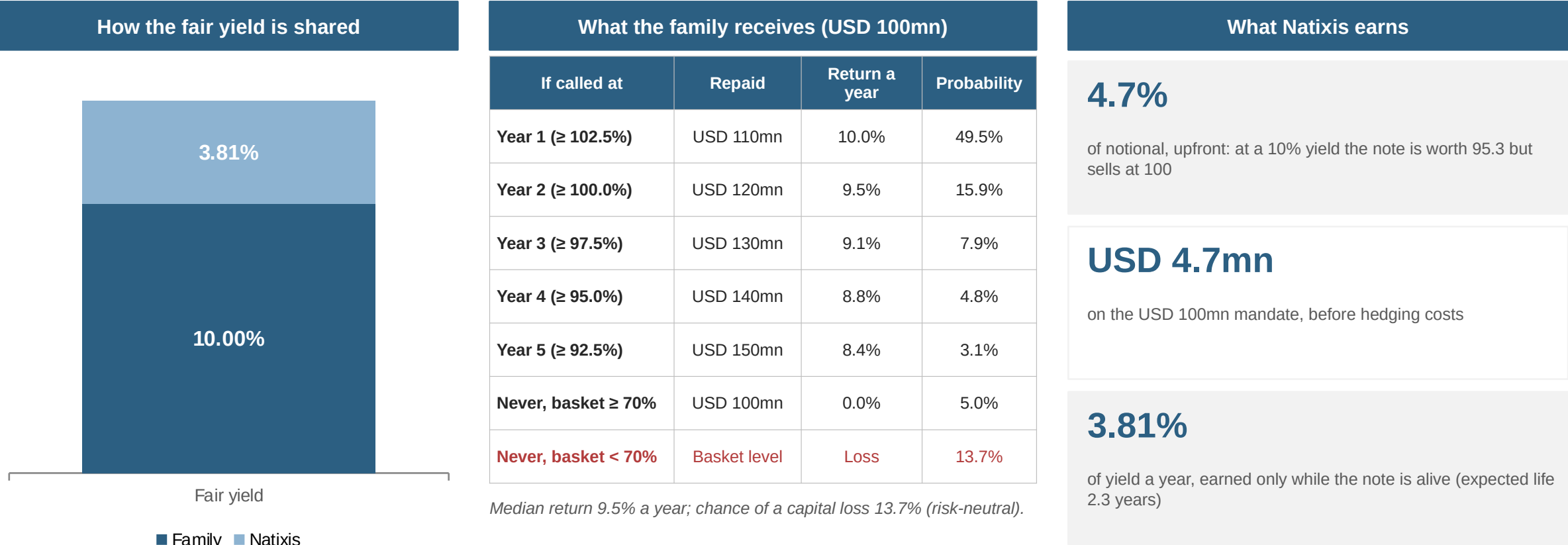
Fair yield	
13.85%	Monte Carlo point estimate (standard error 0.02%)
13.81% – 13.90%	95% confidence interval
13.81%	Fair yield we use: the lower bound, conservative
8.77%	If never-called paths also earned the coupon (alternative Case B)

About half the paths end at Year 1; the coupon is paid for by the 13.7% of paths that end below the 70% barrier

Case A (base): no coupon on never-called paths. Probabilities are risk-neutral (drift = 5.69%), not forecasts. Model: correlated lognormal, deterministic vol term structure, zero dividends.

Splitting 13.81%: 10% a year to the family, 3.81% to Natixis

The family is offered a 10% yield; the gap to the fair yield is Natixis's hedging cost and profit



The family earns 10% a year in the 81.2% of paths that are called; Natixis keeps 4.7% of notional for hedging and margin

Natixis value = fair value at 13.81% (100%) minus fair value at 10% (95.3%). Return a year = $(1 + 10\% \times k)^{(1/k)} - 1$. Excludes issuer credit and hedging costs, which come out of Natixis's share.